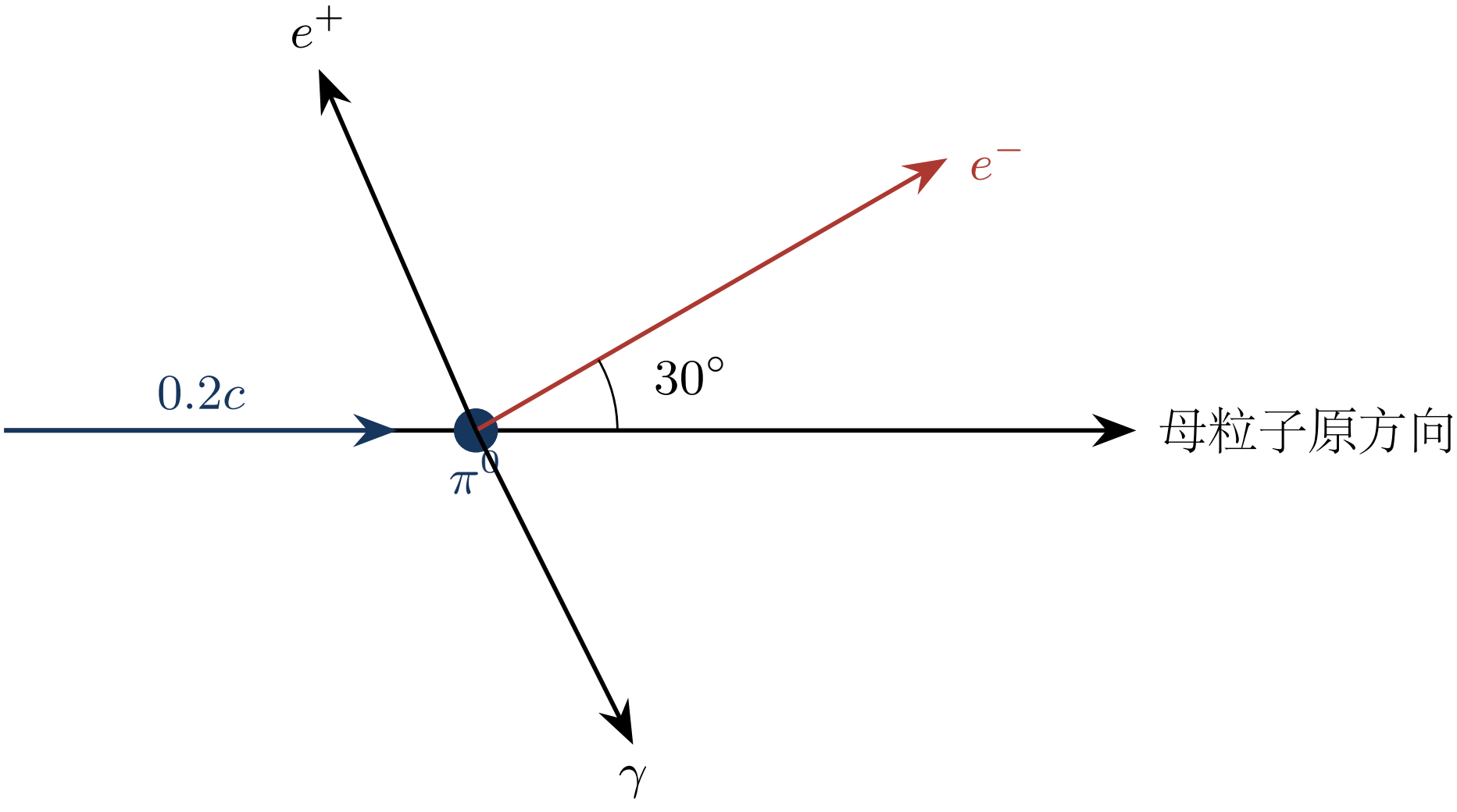
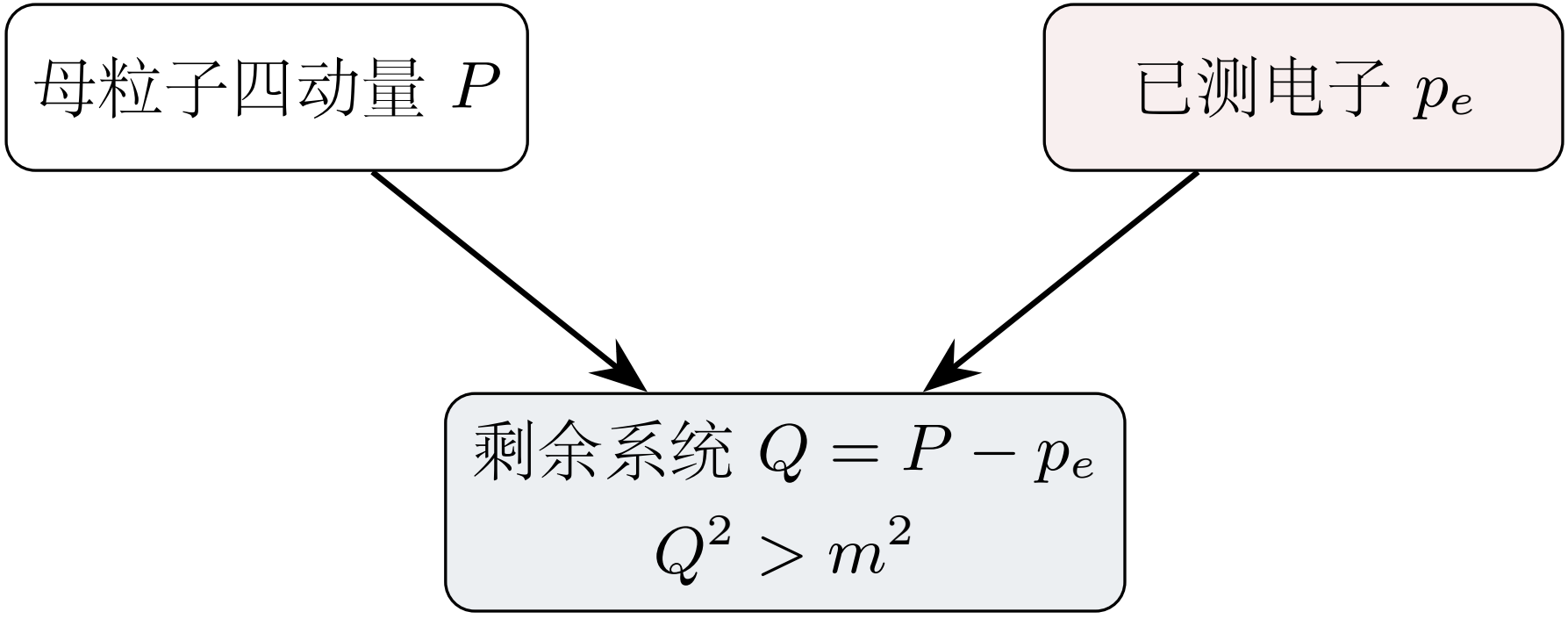
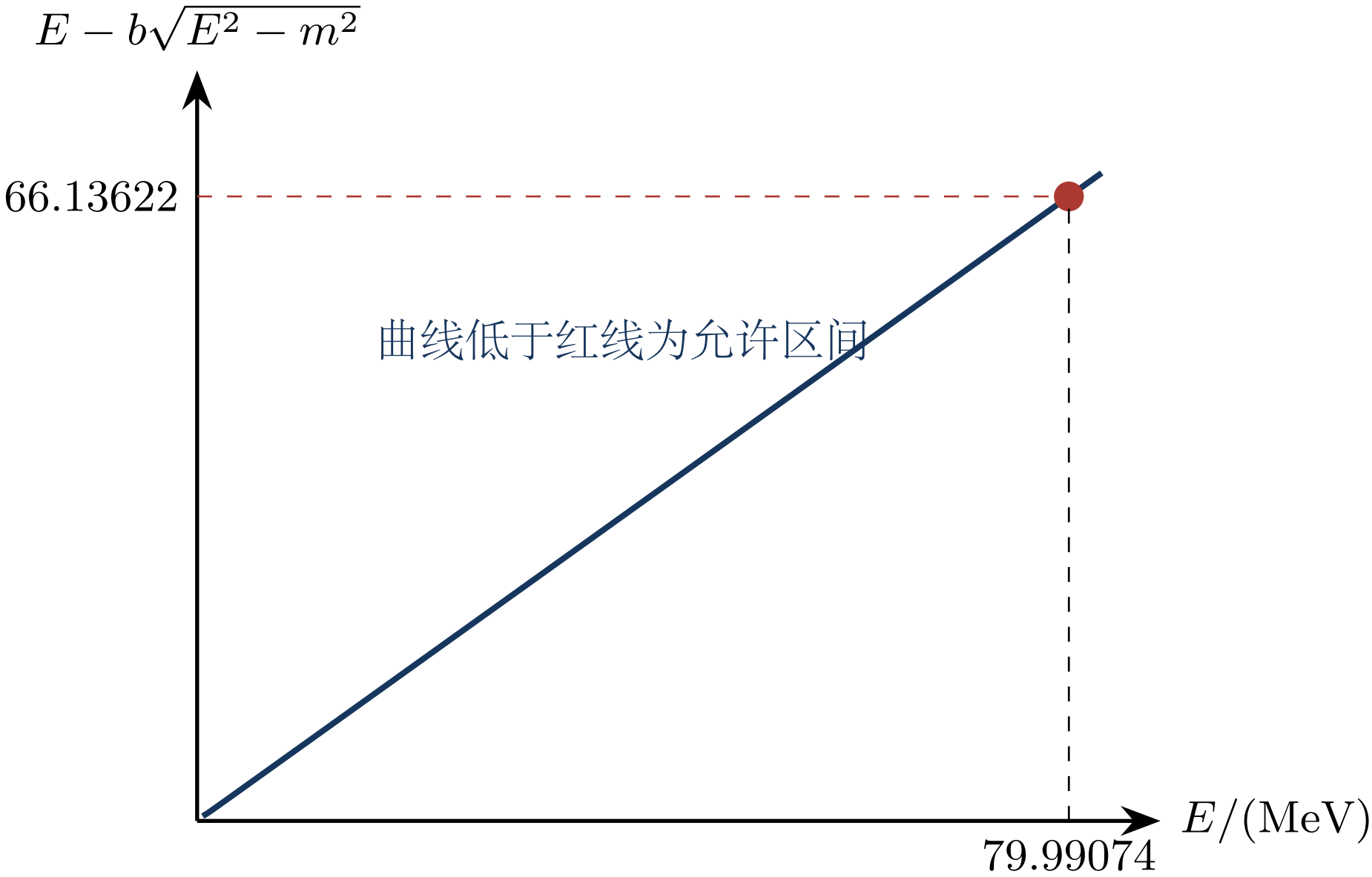
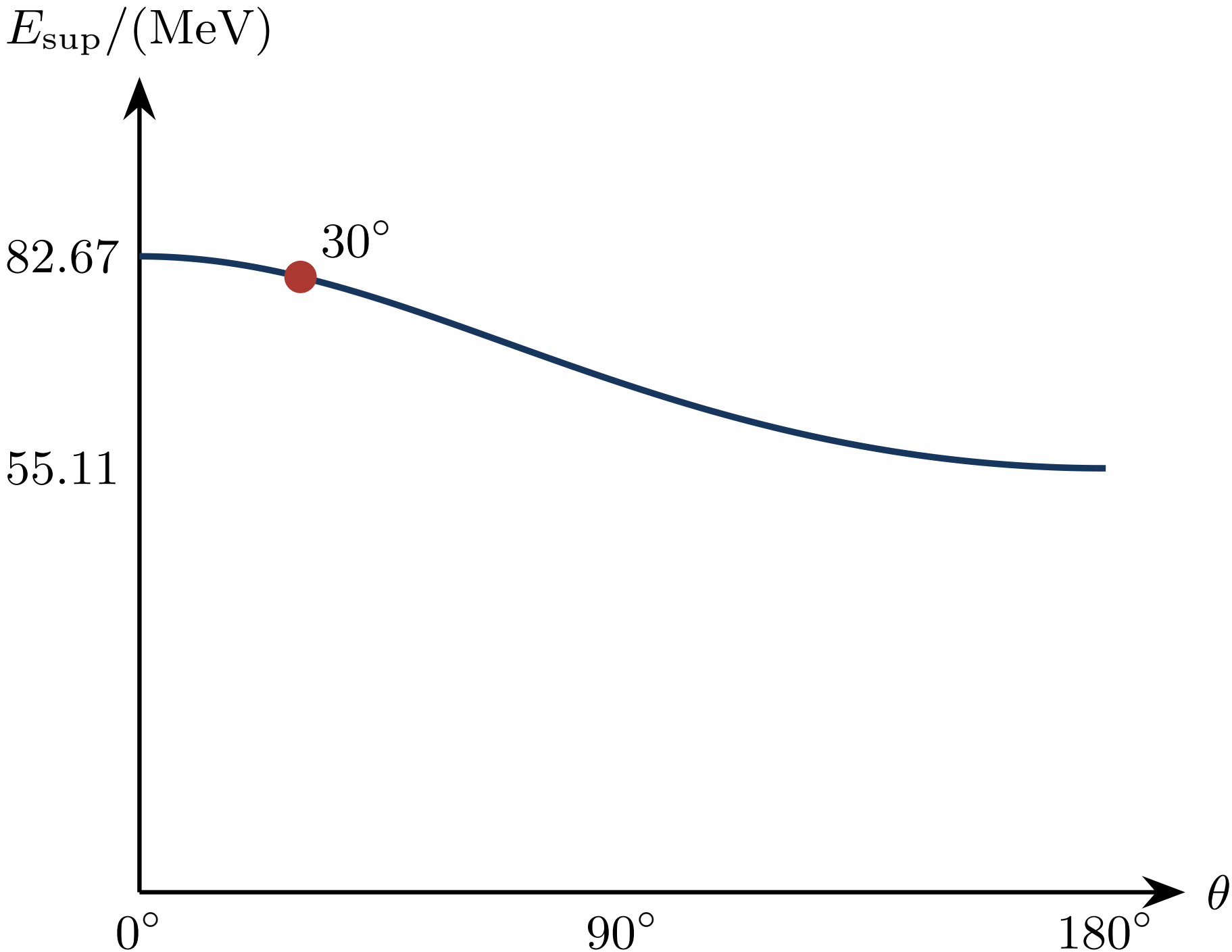


未观测的两条方向仅为示意



非零光子使不变质量严格大于 m


$E - b\sqrt{E^2 - m^2}$
66.13622
曲线低于红线为允许区间
$E/(\mathrm{MeV})$
79.99074


$E_{\mathrm{sup}}/(\mathrm{MeV})$
82.67
55.11
30°
0°
90°
180°
θ

下确界
方向不定义
上确界
光子能量趋于零
$v = 0$
$v = 0.999979595c$
真实有向三体事件：严格开区间